Anteater Chess v1.0

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Table of Contents

Glossary (of chess terms) **2**
1 Computer Chess **3**
1.1 Usage Scenario 3
1.2 Goals 3
1.3 Features 3
2 Installation **4**
2.1 System Requirements 4
2.2 Setup and Configuration 4
2.3 Uninstalling 4
3 Chess Program Functions and Features **5**
3.1 Starting a New Game 5
3.2 Printing the Chess Board 5
3.3 Making a Move 5
3.4 Anteater Piece and Ant Eating 5
3.5 Special Chess Moves 5
3.6 Check and Checkmate Detection 5
3.7 Move Log 5

Glossary (of chess terms)

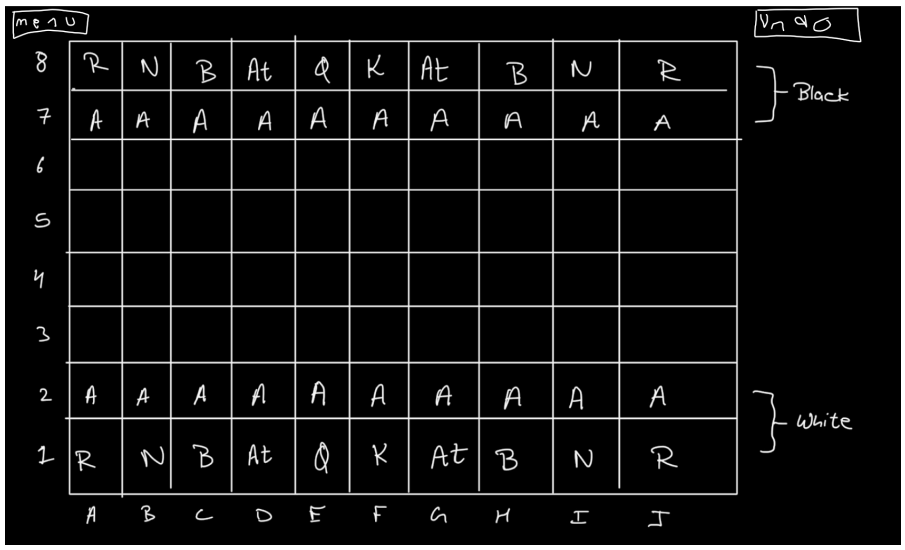
Term	Definition
Ant	An alternate name for a pawn in Anteater Chess.
Ant Eating	A special move unique to Anteater Chess where the anteater piece captures one or more adjacent pawns (ants) in a single move.
Anteater	A new chess piece unique to Anteater Chess. It moves one step in any direction but can only capture ants (pawns), potentially eating multiple ants in one move.
Bishop	A classic chess piece that moves diagonally any number of squares.
Board	The 10×8 grid on which Anteater Chess is played, with columns labeled A–J and rows labeled 1–8.
Castling	A special move involving the king and a rook, performed under specific conditions.
Check	A situation in which a player's king is under threat of capture on the opponent's next move.
Checkmate	A game-ending situation in which a player's king is in check and has no legal move to escape.
Computer Player	The automated opponent controlled by the Anteater Chess program.
En Passant	A special pawn capture move that can occur immediately after an opponent moves a pawn two squares forward.
File	A vertical column on the chess board, labeled A through J.
King	The most important chess piece; the game ends when the king is checkmated.
Knight	A chess piece that moves in an L-shape: two squares in one direction and one square perpendicular.
Legal Move	A move that conforms to all rules of Anteater Chess.
Move Log	A human-readable text file recording all moves made during a game.
Pawn	A chess piece (also called an ant) that moves forward one square at a time (two from the starting position) and captures diagonally.
Promotion	The transformation of a pawn into another piece (usually a queen/user's choice) upon reaching the far end of the board.
Queen	The most powerful classic chess piece; moves horizontally, vertically, or diagonally any number of squares.
Rank	A horizontal row on the chess board, numbered 1 through 8.
Rook	A chess piece that moves horizontally or vertically any number of squares.
Square	A single cell on the chess board, identified by a letter (file) and number (rank), e.g., F4.
Stalemate	A game-ending situation in which a player has no legal move available to them and they are not in a check/checkmate. Can also occur when there is insufficient material for both sides (but the latter is more complicated).

1 Computer Chess

1.1 Usage Scenario

Anteater Chess is a two-player strategy game. Upon launching the program, the player can select the game mode: single player game (versus a computer) or a two player game(played with another human in person). Once this is selected the player or players can choose their side (white or black) and the game begins. The graphical user interface displays a 10×8 chess board with all pieces in their starting positions. The player makes moves by selecting a piece and its destination square; the computer then responds with its own move. The game continues until one player achieves checkmate. The GUI also includes an undo and menu button. The undo button undoes the most recent move made by the human player and the menu button provides options to quit, start a new game, and resume. Prior to the GUI, the user can undo, quit, or start new game via command line arguments.

Below is the sketch of a typical game screen, before the GUI is available.



1.2 Goals

The primary goal of Anteater Chess is to checkmate the opponent's king – that is, to put the king in a position where it is under attack (in check) and has no legal move to escape. The game is lost by the player whose king is checkmated.

The goals of the Anteater Chess software program are:

- Provide a complete and rule-compliant implementation of Anteater Chess, including all standard chess rules and the Anteater extensions.
- Offer an intuitive and accessible user interface for players of all skill levels.
- Deliver a capable computer opponent that responds within a reasonable time (under one minute per move).
- Log all game moves to a human-readable text file for review.

- Serve as a platform for competitive play in the EECS 22L Anteater Chess Tournament.

1.3 Features

Anteater Chess includes the following key features:

Core Features:

- Full implementation of Anteater Chess rules including all classic chess moves and the unique anteater piece and ant-eating mechanic.
- Graphical User Interface (GUI) displaying the board and pieces in their current positions.
- Text-based (ASCII) board display as the initial interface.
- Interactive human player vs. computer player gameplay.
- Player choice of side: white or black.
- Automated move logging to a text file.
- Human vs. Human and Computer vs. Computer modes.
- Multiple difficulty levels: beginner, intermediate, and expert.
- Save and load game state.

Optional / Bonus Features:

- Undo (withdraw) previous move functionality.
- Move hints for the human player.

Game clocks and timers for each player.

2 Installation

2.1 System Requirements

To ensure Anteater Chess program will run correctly on your device, the following requirements must be met:

- Operating System: Windows, macOS Ventura or later, or Linux
- Processor: Intel I3 or equivalent (minimum)
- Memory: 4 GB minimum
- Storage: At least 200 MB of free disk space
- Others: GCC Compiler
- Internet: For updates (optional)

2.2 Setup and Configuration

Please refer to these instructions when installing Anteater Chess:

Step 1: Download Program

Obtain source code
- Can be done through Canvas submission files

Step 2: Extract Files

If the program file is compressed, extract it using the following:

```
unzip anteater_chess.zip
cd anteater_chess
```

Step 3: Compile the Program

Use the following makefile to generate executable file “AnteaterChess”:

```
make AnteaterChess
```

Step 4: Run the Program

Run the following line to start the program:

```
./AnteaterChess
```

Step 5: Follow the Directions on Screen

To start a game:
1. Choose your difficulty level(Easy/Medium/Hard)

2. Pick the side you wanna play as(Black/White)
3. Select if you want to play vs AI or Human
4. Hit Enter to Start

2.3 Uninstalling

Please refer to these instructions when uninstalling chess:

Run the following in terminal:

1. Reach installation directory using the following command:

```
cd anteater_chess
```

2. Remove all files using:

```
rm -rf anteather_chess
```

For Windows:

Right click and press “Delete” on Windows program

3 Chess Program Functions and Features

3.1 Starting a New Game

User Input: The player launches the program from the command line. Upon startup, the program presents a menu with options to start a new game, load a saved game, or exit.

Program Output: The program displays a welcome screen followed by a prompt asking the player to choose a color (white or black). After the player selects a color, the board is initialized to the starting position and displayed on screen.

Example interaction:

```
Welcome to Anteater Chess v1.0
==============================
[1] New Game
[2] Load Game
[3] Exit
Enter choice: 1
Choose your color [w/b]: w
You play as White. Computer plays as Black.
Press ENTER to begin...
```

After selection, the board is rendered (text or graphical mode) with white pieces at rows 1-2 and black pieces at rows 7-8. The program notifies the player whose turn it is, and waits for input.

3.2 Displaying the Chess Board

User Input: No explicit user input is required to trigger a board display. The board is automatically printed to the screen after every move (both human and computer), at the start of the game, and whenever the board state changes.

Program Output: The program renders the current state of the Anteater Chess board. Two display modes are supported:

```
  +----+----+----+----+----+----+----+----+----+----+
8 | bR | bN | bB | bA | bQ | bK | bA | bB | bN | bR |
  +----+----+----+----+----+----+----+----+----+----+
7 | bP | bP | bP | bP | bP | bP | bP | bP | bP | bP |
  +----+----+----+----+----+----+----+----+----+----+
6 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
```

```
5 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
4 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
3 |    |    |    |    |    |    |    |    |    |    |
  +----+----+----+----+----+----+----+----+----+----+
2 | wP | wP | wP | wP | wP | wP | wP | wP | wP | wP |
  +----+----+----+----+----+----+----+----+----+----+
1 | wR | wN | wB | wA | wQ | wK | wA | wB | wN | wR |
  +----+----+----+----+----+----+----+----+----+----+
    A    B    C    D    E    F    G    H    I    J
```

Text / ASCII Mode (Initial Interface)

The board is drawn using ASCII characters. Each cell is a 4-character wide slot. Pieces are shown with a two-letter code: the first letter indicates color (w = white, b = black) and the second indicates the piece type (R=Rook, N=Knight, B=Bishop, Q=Queen, K=King, A=Anteater, P=Pawn). Empty squares are shown as blank.

3.3 Making a Move

- ➔ Queen can move horizontally, vertically or diagonally any number of spaces across the board
- ➔ Rook can move any number of spaces horizontally or vertically in a straight line across the board
- ➔ A knight has to travel in an L-shape (first three straight spaces vertically or horizontally). Then after travelling three spaces, the knight has to move 2 spaces perpendicular to the space it is in. The 2 spaces can be traveled in either direction, but it should be perpendicular to the direction of its first 3 spaces it moved.
- ➔ The king can move in any direction, but only one space at a time. A king can also never step into the space for check and must avoid getting into the space where it can be killed by an enemy piece during check.
- ➔ The ant can only move one space forward at a time, when not killing or capturing an enemy piece. To capture an enemy piece, it may move one space diagonal only if the enemy piece is in a diagonal space in front of the team's pawn.
- ➔ An anteater can move in any direction, but only one space at a time, just like a king. An anteater can only kill pawns (ants), and it may capture many ants in one move. When capturing multiple ants at once, the anteater can eat adjacent ants in its one move. This will be explained more in *ant eating,* a move listed in special moves.

3.4 Anteater Pieces and Ant Eating

The anteater is the unique piece introduced in Anteater Chess. It behaves differently from all classic pieces and requires special handling by the program.

User Input: To move the anteater, the player enters the source square (where the anteater currently is) and the destination square. If the destination is occupied by an ant (pawn) and additional adjacent ants are present, the program will prompt the player to specify how far to eat along the row or column.

Program Output: The program validates anteater moves according to the following rules:

- The anteater can move one step in any direction (like a king), provided the destination is empty or occupied by an enemy ant.
- The anteater cannot capture any piece other than ants.
- When the anteater captures an ant, it may also eat additional adjacent ants along the same file or rank in the same move (ant eating).
- The final position of the anteater after ant eating is the square of the last ant eaten, as chosen by the player.
- The anteater is not a threat to the opposing king and cannot put a king in check.

Example: If the anteater is at D6 and ants are at E7, F7, G7, and G6, the player may move the anteater from D6 to G6 by eating the ants at E7, F7, G7, and G6 in sequence. The ant at H7 would remain untouched.

```
White's turn. Enter move: D6 G6
Anteater eats ants at E7, F7, G7, G6.
Anteater moves to G6.
[board updated]
```

3.5 Special Chess Moves

3.5.1 Castling:

Castling is the only move in this game where two pieces move at once. It involves the king and either rook of the same player. This move is only viable if neither the king nor the rook has previously made a move, and all the squares are unoccupied. Some other conditions include that the king is not in check and the king doesn't end up in an attacked square by the opponent's piece. The castle move involves the king moving 2 squares towards the rook and the rook jumping to the king's original square. If the board is a 10x10 the castling position is adjusted according to the standard move on the 8x8 board.

3.5.2 En Passant:

En passant prevents a pawn from bypassing an enemy pawn using its two square opening. The pawn advances two squares from its starting position and the enemy pawn occupies an adjacent square on the same rank. The enemy pawn captures diagonally to the square the attacking pawn passed through. The advancing pawn is removed even though the capturing pawn does not occupy its square. This means the only move in chess where capture

can happen without landing on the captured pieces square. En passant must be executed on the very next turn or the right is permanently lost. If it's the only legal move it must be made.

3.5.3 Pawn Promotion:

When a pawn from either player reaches the opposite far end of the board, rank 8 for white and rank 1 for black, it has the option to be promoted to a Queen, rook, Bishop or knight of the same color, chosen by the player. The program automatically promotes the pawn to a queen if no selection is made. The promoted piece is active immediately and may deliver a check on the same turn.

3.5.4 ANTEATING

The anteaters special capture mechanic allows the piece to consume multiple enemy pawns in a single move. The anteater just like the king can only move one square at a time in any direction. It only captures enemy pawns and can't threaten any other piece including the opponent's king. When an enemy pawn is adjacent the anteater may eat it. After eating it if there's another pawn adjacent to the next position the anteater may continue eating it in the same move. The player chooses how far the anteater travels. It does not need to eat every reachable ant. The player chooses how far the anteater travels. It doesn't need to eat every reachable ant. The anteater ends its move at the position of the last ant eaten.

3.6 Check and Checkmate Detection

3.6.1 Check

A king is in check when it is threatened by at least one enemy piece. After every move, the program scans all enemy pieces to determine whether the active player's king can be reached. If so a check notification is displayed and the player must resolve it immediately no other move is permitted. The possible moves include moving a king to a square not attached by any enemy piece, using another piece to block the attack or capturing the attacking piece that is threatening the king.

3.6.2 Checkmate

Checkmate occurs when the king is in check and no legal move can escape it. The game ends immediately. To detect a checkmate the program must confirm the current player's king in check, generate every possible move for every friendly piece, each move is simulated on a copy of the board, and the king remains in check even after every candidate move, checkmate is declared and the player loses.

3.6.3 Stalemate

Stalemate occurs when the current player is not in check but has no legal moves available. The result is a draw. The program detects this condition and declares a stalemate before ending the game.

3.7 Move Log

The program maintains a running move log displayed alongside the board during gameplay. Each move is recorded in standard algebraic notation (e.g., e4, Nf3, O-O for castling) and numbered by turn. Special moves are annotated accordingly — en passant is marked with "e.p.", pawn promotion includes the promoted piece (e.g., e8=Q), and anteater ant-eating sequences are logged with each consumed pawn's square listed in order.

User Input: The player may enter a command (e.g., "log") to view the full move history at any time.

Program Output: The move log is displayed as a two-column list pairing White's and Black's moves per turn. The log scrolls if the game exceeds the visible display area. At the end of the game, the full log is shown and the player is given the option to save it to a text file.

Example:

```
Move Log
========
1.  e4        e5
2.  Nf3       Nc6
3.  D6-G6*    Bb5
    (Anteater eats E7, F7, G7, G6)
```

3.8 Computer Player (AI Opponent)

The computer opponent controls the opposing color selected by the player at game start. The AI evaluates the board state each turn and selects a legal move using a minimax algorithm with alpha-beta pruning to a set search depth.

Program Output: After the player submits a move, the program displays a brief "Computer is thinking..." message, then outputs the AI's chosen move in algebraic notation and updates the board. The AI is capable of executing all legal moves including castling, en passant, pawn promotion, and anteater ant-eating sequences. The AI will default to promoting pawns to queens unless a different piece yields a clearly superior position.

Difficulty is fixed for the base implementation but the program architecture should allow for depth adjustment in future versions.

```
Computer is thinking...
Computer plays: Nf6
[board updated]
```

3.9 Save and Load Game

3.9.1 Saving

Saving: The player may save the current game state at any time by entering a save command (e.g., "save"). The program writes the full game state to a file, including the current board position, move log, whose turn it is, castling rights, en passant eligibility, and any anteater-specific state. The player is prompted to enter a filename; if none is provided, the program generates a default name using a timestamp (e.g., anteatarchess_2025-04-05.sav).

Program Output:

```
Game saved as: anteatarchess_2025-04-05.sav
```

3.9.2 Loading

Loading: From the main menu, the player selects option [2] Load Game. The program lists all available save files in the current directory and prompts the player to select one. Once loaded, the board is restored exactly to the saved state and the game resumes from that point.

Example interaction:

```
Available saves:
  [1] anteatarchess_2025-04-05.sav
  [2] midgame_test.sav
Select save file: 1
Game loaded. White's turn.
[board displayed]
```

If a save file is corrupted or incompatible, the program displays an error message and returns to the main menu without crashing.

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The rules of standard chess are in the public domain. The Anteater Chess variant rules are defined by Prof. Rainer Dömer for EECS 22L, UCI.

Error Messages

Error Message	Cause	Recommended Action
`Invalid move. Please try again.`	The entered move is not legal under Anteater Chess rules.	Check the selected piece's legal moves and re-enter a valid move.
`No piece at source square.`	The source square specified does not contain any piece.	Verify the source square and re-enter the move.
`That piece does not belong to you.`	The player attempted to move an opponent's piece.	Select one of your own pieces and re-enter the move.
`Move would put your king in check.`	The attempted move would leave or place the player's king under attack.	Select a different move that protects the king.
`Anteater cannot capture that piece.`	The anteater was directed to capture a non-pawn piece.	Move the anteater to an empty square or to a square with an enemy pawn.
`Illegal castling. Conditions not met.`	Castling was attempted but one or more conditions are not satisfied.	Ensure neither king nor rook has moved, the path is clear, and king is not in check.
`En passant no longer available.`	An en passant capture was attempted but not performed immediately after the qualifying pawn move.	Continue with a standard move. En passant must be done on the very next turn.
`Cannot load file: file not found.`	The specified save file does not exist in the current directory.	Check the filename and path, then try again.
`Error writing log file.`	The move log file could not be written (e.g., disk full, permission denied).	Check available disk space and directory write permissions.
`Display initialization failed.`	The graphical interface could not start (missing library or display).	Verify GUI libraries are installed; fall back to text mode if necessary.
`Unknown command. Type 'help' for options.`	An unrecognized command or input was entered.	Type 'help' to see available commands, or enter a valid move.

Index

AI opponent (computer) …… 9

Board ……..……. 3

Checkmate ……..…… 9

Controls ……..……. 7

Features ……,…..…… 4

Installation ……...……. 5

Move validation …….…..……. 7

New game ……..…….. 7

Settings …….…… 7

Special Moves …….. 8

Stalemate …….. 9

Undo …….……...….. 7

Uninstalling …….…….. 6

User Interface …….…….. 7